ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2024 P2HR Worked Solutions

Jun 2024 | P2HR

33 questions ▪ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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1 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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WORKED SOLUTION

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1. Find the LCM

$$1400 = 14 \times 100$$

$$1400 = (2 \times 7)(2^2 \times 5^2)$$

$$1400 = 2^3 \times 5^2 \times 7$$

FINAL ANSWER

$2^3 \times 5^2 \times 7$

PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Transformations**

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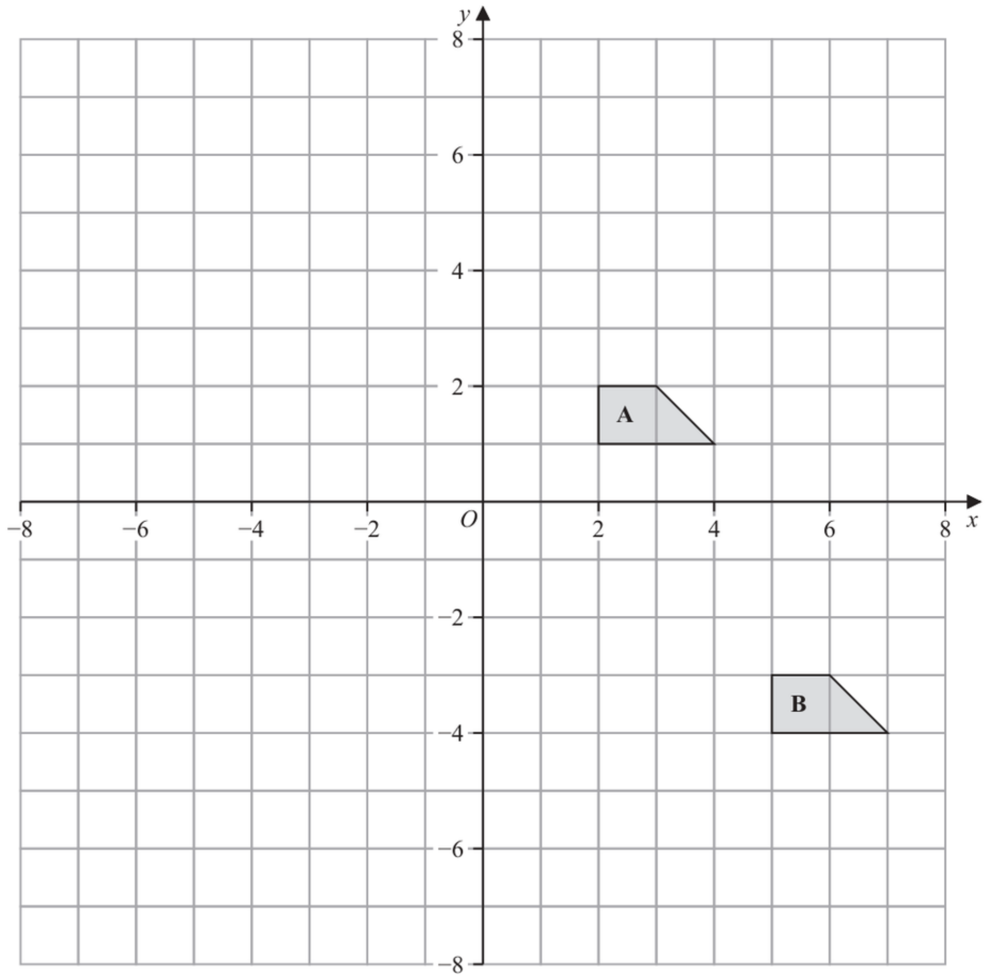
2



(a) Describe fully the single transformation that maps shape **A** onto shape **B**

....

....

(2)

(b) On the grid above, rotate shape **A** 180° about (−1, 0)
Label your shape **C**

(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Transformations**

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WORKED SOLUTION

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1. (a) Shape moves to shape

(a) Shape A moves to shape B by the vector

$$\begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

2. Use matrices

So the transformation is a translation by $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$.

3. (b) For a rotation about

(b) For a 180° rotation about $(-1, 0)$,

$$(x, y) \mapsto (-2 - x,\ -y)$$

4. The image has vertices

The image has vertices

$$(-4, -1),\ (-6, -1),\ (-5, -2),\ (-4, -2)$$

FINAL ANSWER

(a) translation by $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$; (b) vertices $(-4, -1), (-6, -1), (-5, -2), (-4, -2)$.

PAST PAPER QUESTION Q#: 3 Marks: 2

TOPIC: **Statistics Toolkit**

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3 Here is a list of four numbers written in ascending order of size

$$x \qquad x \qquad y \qquad 15$$

where x and y are integers.

The numbers have

a median of 12.5
a range of 4

Find the value of x and the value of y

x = ...

y = ...

(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 2

TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

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1. The range is

The range is 4:

$$15 - x = 4$$

$$x = 11$$

2. The median is

The median is 12.5:

$$\frac{x+y}{2} = 12.5$$

$$x + y = 25$$

$$y = 25 - 11 = 14$$

FINAL ANSWER

$x = 11, \ y = 14.$

PAST PAPER QUESTION

Q#: 4 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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4 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
$A = \{\text{factors of } 6\}$
$B = \{\text{prime numbers}\}$

(a) List the members of the set

(i) $A \cup B$

....

(1)

(ii) A'

....

(1)

Harpreet states that $A \cap B = \varnothing$

Harpreet is incorrect.

(b) Explain why.

....

(1)

C is a set with 4 members such that

the set $A \cap C$ has 2 members
the set $B \cap C$ has 2 members

Set $A \cap C$ and set $B \cap C$ have no members in common.

(c) List the 4 members of set C

....

(2)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION Q#: 4 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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WORKED SOLUTION

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1. Calculate value

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 3, 6\}$$

$$B = \{2, 3, 5, 7\}$$

2. Calculate value

So

$$A \cup B = \{1, 2, 3, 5, 6, 7\}$$

3. Calculate value

and

$$A' = \{4, 5, 7, 8, 9, 10\}$$

4. Harpreet is not correct because

Harpreet is not correct because 2 and 3 are in both A and B.

$$A \cap B = \{2, 3\}$$

5. Calculate value

For C, it has 4 members, $n(A \cap C) = 2$, $n(B \cap C) = 2$, and no member is in both of those overlaps. So choose the two elements of A only and the two elements of B only:

$$C = \{1, 5, 6, 7\}$$

FINAL ANSWER

(a)(i) $\{1, 2, 3, 5, 6, 7\}$, (a)(ii) $\{4, 5, 7, 8, 9, 10\}$, (b) Harpreet is incorrect, (c) $C = \{1, 5, 6, 7\}$

PAST PAPER QUESTION

Q#: 5 Marks: 4

Topic: **Circles, Arcs & Sectors**

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5 The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square *ABCD*

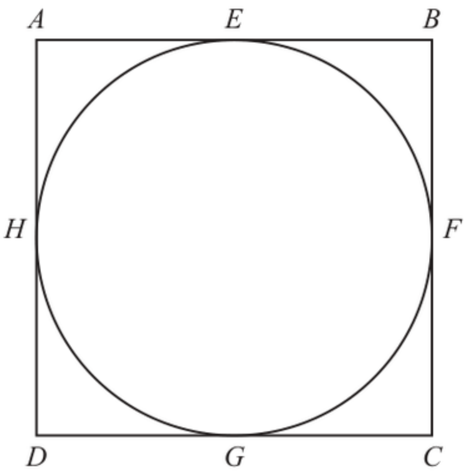

Diagram **NOT** accurately drawn

The circle touches the square at the points *E*, *F*, *G* and *H*

The area of the square is $81\,\text{cm}^2$

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

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WORKED SOLUTION

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1. Calculate area

The square has area 81 cm^2, so its side length is

$$\sqrt{81} = 9 \text{ cm}$$

2. Perimeter of the square

Perimeter of the square:

$$4 \times 9 = 36$$

3. Calculate value

The circle has diameter 9 cm, so its circumference is

$$\pi d = 9\pi$$

4. Total wire length

Total wire length:

$$36 + 9\pi = 64.274\ldots$$

FINAL ANSWER

64.3 cm to 3 significant figures.

PAST PAPER QUESTION

Q#: 6 Marks: 8

TOPIC: **Solving Linear Equations**

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6 (a) Solve $\frac{2f}{3} = 4f - 17$

Show clear algebraic working.

$f =$ ....

(3)

(b) Simplify $(e + 12)^0$ where $e > 0$

....

(1)

(c) Simplify fully $\frac{12a^4h^6}{4ah^2}$

....

(2)

(d) Factorise fully $20x^5y + 12x^3y^4$

....

(2)

(Total for Question 6 is 8 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 8

TOPIC: **Solving Linear Equations**

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WORKED SOLUTION

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1. (a)

(a)

$$\frac{2f}{3} = 4f - 17$$

$$2f = 12f - 51$$

$$10f = 51$$

$$f = \frac{51}{10}$$

2. (b)

(b) $(e + 12)^0 = 1$.

3. (c)

(c)

$$\frac{12a^4h^6}{4ah^2} = 3a^3h^4$$

4. (d)

(d)

$$20x^5y + 12x^3y^4 = 4x^3y(5x^2 + 3y^3)$$

FINAL ANSWER

$f = \frac{51}{10}$, 1, $3a^3h^4$, $4x^3y(5x^2 + 3y^3)$

PAST PAPER QUESTION

Q#: 7 Marks: 2

TOPIC: **Algebraic Roots & Indices**

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7 $\dfrac{3^{-2} \times 3^{5}}{3^{10}} = 3^{n}$

Find the value of n

$n =$ ....

(Total for Question 7 is 2 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 2

TOPIC: **Algebraic Roots & Indices**

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WORKED SOLUTION

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1. Use index laws

$$\frac{3^{-2} \times 3^5}{3^{10}} = 3^n$$

2. Use index laws

Use index laws:

$$\frac{3^{-2+5}}{3^{10}} = 3^{3-10} = 3^{-7}$$

3. Use index laws

So

$$n = -7$$

FINAL ANSWER

$n = -7$.

PAST PAPER QUESTION

Q#: 8 Marks: 3

TOPIC: **Percentages**

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8 In a sale, all normal prices are reduced by 17%

The sale price of a fridge is 6225 rupees.

Work out the normal price of the fridge.

.... rupees

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION Q#: 8 Marks: 3

TOPIC: **Percentages**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

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1. A reduction of means the

A reduction of 17% means the sale price is 83% of the normal price.

$$0.83 \times \text{normal price} = 6225$$

$$\text{normal price} = \frac{6225}{0.83} = 7500$$

FINAL ANSWER

7500 rupees

PAST PAPER QUESTION

Q#: 9 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jun 2024 P2HR - Jun 2024 | P2HR

9 (a) Write 6.04×10^5 as an ordinary number.

....
(1)

(b) Write 0.000 07 in standard form.

....
(1)

(c) Work out $\dfrac{7.6 \times 10^{10}}{4 \times 10^5 - 2 \times 10^4}$

Give your answer in standard form.

....
(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

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WORKED SOLUTION

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1. Convert standard form

$$6.04 \times 10^5 = 604000$$

$$0.00007 = 7 \times 10^{-5}$$

2. Convert standard form

For part (c), first simplify the denominator:

$$4 \times 10^5 - 2 \times 10^4 = 400000 - 20000 = 380000$$

$$380000 = 3.8 \times 10^5$$

3. Convert standard form

Then

$$\frac{7.6 \times 10^{10}}{3.8 \times 10^5} = 2 \times 10^5$$

FINAL ANSWER

(a) 604000, (b) 7×10^{-5}, (c) 2×10^5

PAST PAPER QUESTION **Q#: 10** **Marks: 5**

TOPIC: **Area & Perimeter**

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10 The diagram shows a hexagon *ABCDEF*

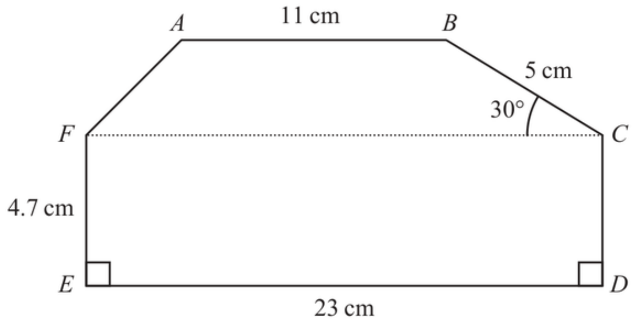

Diagram **NOT** accurately drawn

Angle *BCF* = 30°
AB, *FC* and *ED* are parallel.

Calculate the area of *ABCDEF*
Show your working clearly.

.... cm^2

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

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WORKED SOLUTION

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1. The rectangle has area

The rectangle $EDCF$ has area

$$23 \times 4.7 = 108.1$$

2. The height of the top

The height of the top trapezium is

$$5 \sin 30^\circ = 2.5$$

3. Area of the top trapezium

Area of the top trapezium:

$$\frac{1}{2}(11 + 23) \times 2.5 = 42.5$$

4. Total area

Total area:

$$108.1 + 42.5 = 150.6$$

FINAL ANSWER

150.6 cm^2.

PAST PAPER QUESTION

Q#: 11 Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

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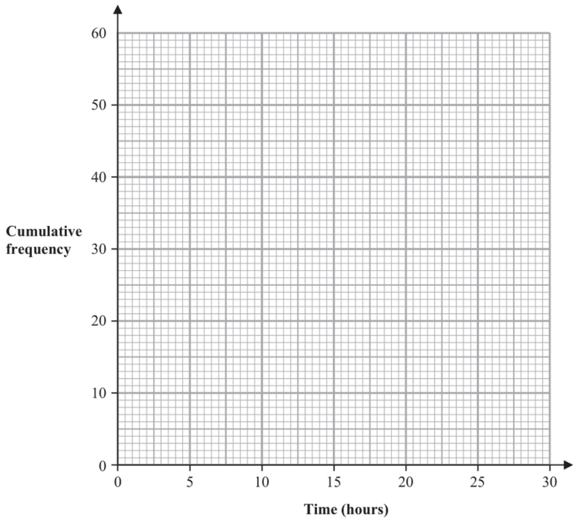
11 The cumulative frequency table gives information about the time, in hours, that each of 60 workers spent working from home in one week.

Time (t hours)	Cumulative frequency
$0 < t \leqslant 5$	6
$0 < t \leqslant 10$	17
$0 < t \leqslant 15$	27
$0 < t \leqslant 20$	42
$0 < t \leqslant 25$	53
$0 < t \leqslant 30$	60

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the interquartile range of the times.

.... hours

(2)

25 workers spent more than W hours working from home.

(c) Use your graph to find an estimate for the value of W

$W =$ ....

(2)

One of the 60 workers is chosen at random.
This worker spent H hours working from home.

(d) Find the probability that $5 < H \leqslant 10$

....

(1)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

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WORKED SOLUTION

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1. The cumulative frequencies are

The cumulative frequencies are

$$6,\ 17,\ 27,\ 42,\ 53,\ 60$$

2. Plot

Plot

$$(0,0),(5,6),(10,17),(15,27),(20,42),(25,53),(30,60)$$

3. Use cumulative frequency

For 60 workers,

$$Q_1 = 15, \qquad Q_3 = 45$$

4. Use cumulative frequency

From the graph,

$$Q_1 \approx 9.1, \qquad Q_3 \approx 21.4$$

$$\text{IQR} \approx 21.4 - 9.1 = 12.3$$

5. Use cumulative frequency

If 25 workers did more than W hours, then 35 workers did W hours or less.
From the graph, $W \approx 17.7$.

$$P(5 < H \leq 10) = \frac{17-6}{60} = \frac{11}{60}$$

FINAL ANSWER

IQR about 12.3, $W \approx 17.7$, and $\frac{11}{60}$.

PAST PAPER QUESTION Q#: 12 Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

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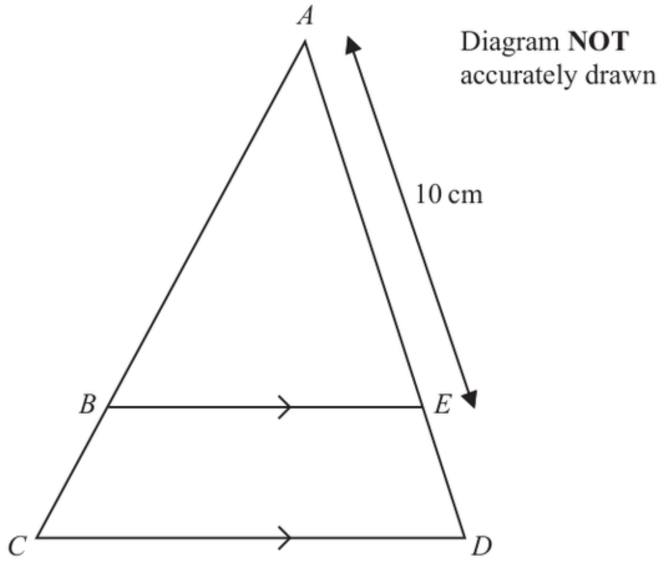
12



In the diagram, ABC and AED are straight lines.
BE is parallel to CD

$AE = 10$ cm and $CD = 1.5 \times BE$

(a) Work out the length of ED

.... cm
(2)

$AB = (2x + 5)$ cm and $BC = (3x - 5)$ cm

(b) Work out the value of x

$x =$ ....
(2)

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

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WORKED SOLUTION

Dr. Eslam Ahmed

1. Find the gradient

Since $BE \parallel CD$, triangles ABE and ACD are similar.

2. Find the gradient

Also

$$CD = 1.5BE$$

so

$$\frac{BE}{CD} = \frac{1}{1.5} = \frac{2}{3}$$

3. Find the gradient

Therefore

$$\frac{AE}{AD} = \frac{2}{3}$$

4. Find the gradient

Given $AE = 10$,

$$\frac{10}{AD} = \frac{2}{3}$$

$$AD = 15$$

5. Find the gradient

So

$$ED = AD - AE = 15 - 10 = 5$$

6. Find the gradient

For part (b),

$$AB = 2x + 5, \qquad BC = 3x - 5$$

$$AC = AB + BC = 5x$$

7. Find the gradient

Using similarity,

$$\frac{AB}{AC} = \frac{2}{3}$$

$$\frac{2x+5}{5x} = \frac{2}{3}$$

$$3(2x+5) = 10x$$

$$6x + 15 = 10x$$

$$x = 3.75$$

FINAL ANSWER
(a) $ED = 5$ cm, (b) $x = 3.75$.

PAST PAPER QUESTION

Q#: 13 Marks: 3

TOPIC: **Circles, Arcs & Sectors**

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13 OAB is a sector of a circle with centre O and radius r cm.

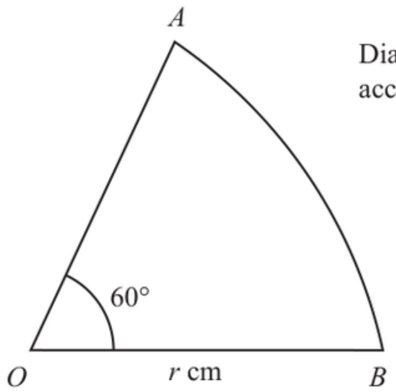

Diagram **NOT** accurately drawn

Angle $AOB = 60°$
The perimeter of the sector is P cm.

Find a formula for P in terms of r
Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Circles, Arcs & Sectors**

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WORKED SOLUTION

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1. The arc length is

The arc length is

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

2. The perimeter is two radii

The perimeter is two radii plus the arc:

$$P = 2r + \frac{\pi r}{3}$$

$$P = r\left(\frac{\pi}{3} + 2\right)$$

FINAL ANSWER

$P = r\left(\frac{1}{3}\pi + 2\right)$.

PAST PAPER QUESTION

Q#: 14 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2024 P2HR - Jun 2024 | P2HR

14 Adriana is going to roll a biased dice and spin a biased coin.

The probability that the coin will land on Heads is 0.8
The probability that the dice will land on 6 and the coin will land on Heads is 0.24

Work out the probability that the dice will land on 6 and the coin will land on Tails.

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION Q#: 14 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate probability

$$P(H) = 0.8$$

2. Given

Given

$$P(6 \text{ and H}) = 0.24$$

$$P(6) = 0.24 \div 0.8 = 0.3$$

3. Calculate probability

Also

$$P(T) = 1 - 0.8 = 0.2$$

$$P(6 \text{ and T}) = 0.3 \times 0.2 = 0.06$$

FINAL ANSWER

0.06.

PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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15

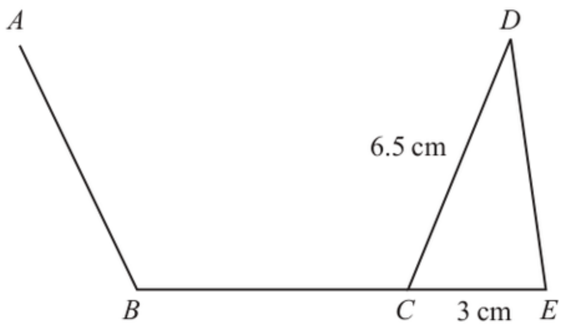

Diagram **NOT** accurately drawn

AB, *BC* and *CD* are three sides of a regular pentagon and *CDE* is a triangle.
BCE is a straight line.

$CD = 6.5\,\text{cm} \qquad CE = 3\,\text{cm}$

Work out the area of triangle *CDE*
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION Q#: 15 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. A regular pentagon has interior

A regular pentagon has interior angle

$$\frac{(5-2)\times 180}{5} = 108^\circ$$

2. Use trigonometry

Since BCE is a straight line,

$$\angle DCE = 180^\circ - 108^\circ = 72^\circ$$

3. Area of triangle is

Area of triangle CDE is

$$\frac{1}{2}ab\sin C$$

$$\text{Area} = \frac{1}{2}(6.5)(3)\sin 72^\circ$$

$$\text{Area} = 9.27\ldots$$

FINAL ANSWER

9.27 cm^2 to 3 significant figures.

PAST PAPER QUESTION

Q#: 16 Marks: 2

TOPIC: **Graphs of Functions**

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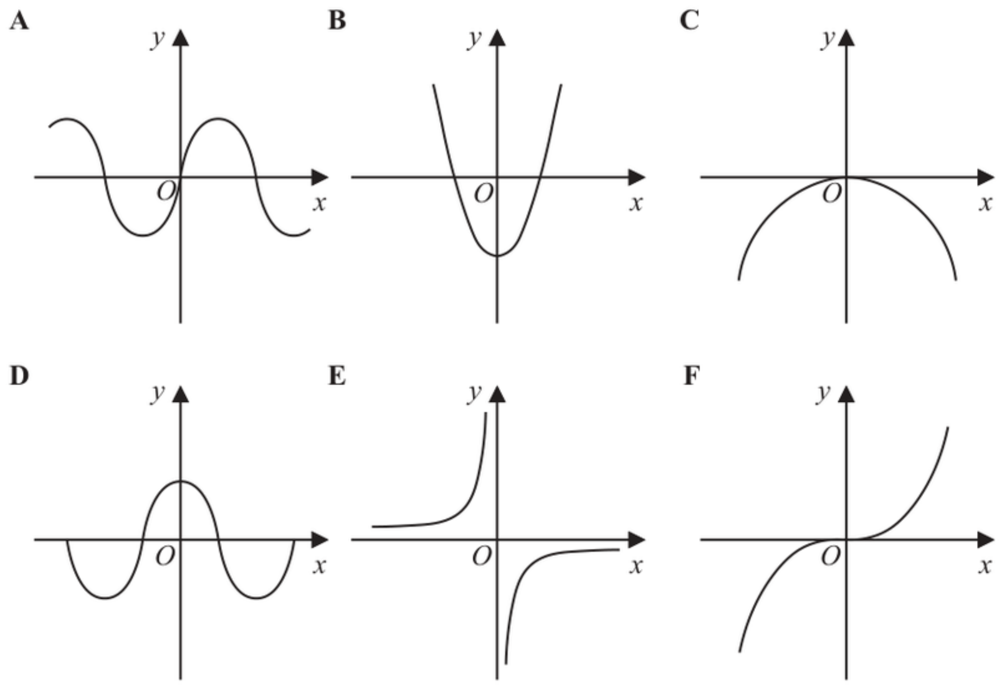
16 Here are six graphs.



Write down the letter of the graph that could have the equation

(i) $y = -\frac{1}{x}$

....

(1)

(ii) $y = \sin x°$

....

(1)

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 2

TOPIC: **Graphs of Functions**

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WORKED SOLUTION

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1. (i) has branches in quadrants

(i)

$$y = -\frac{1}{x}$$

has branches in quadrants II and IV, so it is **Graph E**.

2. (ii) is the sine shaped

(ii)

$$y = \sin x^\circ$$

is the sine-shaped graph through the origin, so it is **Graph A**.

FINAL ANSWER

(i) E, (ii) A.

PAST PAPER QUESTION Q#: 17 Marks: 3

TOPIC: **Functions**

Jun 2024 P2HR - Jun 2024 | P2HR

17 $f(x) = \dfrac{x}{2x - 4}$ $\qquad g(x) = 3x + 1$

Given that $fg(k) = 2$

work out the value of k

$k =$ ....

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Functions**

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WORKED SOLUTION

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1. Evaluate fraction

$$f(x) = \frac{k}{2x-4}, \qquad g(x) = 3x+1$$

2. Given

Given

$$fg(k) = 2$$

3. This means

This means

$$f(g(k)) = 2$$

4. Calculate value

Now

$$g(k) = 3k+1$$

5. Evaluate fraction

So

$$f(g(k)) = \frac{k}{2(3k+1)-4} = \frac{k}{6k-2}$$

$$\frac{k}{6k-2} = 2$$

$$k = 12k - 4$$

$$11k = 4$$

$$k = \frac{4}{11}$$

FINAL ANSWER

$k = \frac{4}{11}$.

PAST PAPER QUESTION **Q#: 18** **Marks: 2**

TOPIC: **Algebraic Proof**

Jun 2024 P2HR - Jun 2024 | P2HR

18 Use algebra to show that $0.\dot{3}0\dot{6} = \dfrac{34}{111}$

(Total for Question 18 is 2 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 2

TOPIC: **Algebraic Proof**

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WORKED SOLUTION

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1. Let

Let

$$x = 0.\dot{3}0\dot{6}$$

2. Calculate value

Then

$$1000x = 306.\dot{3}0\dot{6}$$

3. Subtract

Subtract:

$$1000x - x = 306$$

$$999x = 306$$

$$x = \frac{306}{999} = \frac{34}{111}$$

FINAL ANSWER

$0.\dot{3}0\dot{6} = \dfrac{34}{111}$.

PAST PAPER QUESTION

Q#: 19 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jun 2024 P2HR - Jun 2024 | P2HR

19 Aviv goes on a cycle journey.

For the cycle journey

average speed = 19 km/h correct to the nearest whole number

time = 1.5 hours correct to one decimal place

Work out the upper bound for the distance Aviv travels.
Give your answer correct to 3 significant figures.

.... km

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

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1. Distance is found using

Distance is found using

$$\text{distance} = \text{speed} \times \text{time}$$

2. Find upper bound

For the upper bound,

$$\text{speed} < 19.5, \quad \text{time} < 1.55$$

$$\text{distance}_{\text{upper}} = 19.5 \times 1.55 = 30.225$$

3. Estimate the value

Correct to 3 significant figures,

$$30.225 \approx 30.2$$

FINAL ANSWER

30.2 km

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Solve quadratic equation

$$6x^2 - 7x - 20 > 0$$

2. Factorise

Factorise:

$$6x^2 - 7x - 20 = (3x + 4)(2x - 5)$$

3. The critical values are

The critical values are

$$x = -\frac{4}{3} \quad \text{and} \quad x = \frac{5}{2}$$

4. Solve quadratic equation

Since the coefficient of x^2 is positive, the quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -\frac{4}{3}$ or $x > \frac{5}{2}$

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Solve quadratic equation

$$6x^2 - 7x - 20 > 0$$

2. Factorise

Factorise:

$$6x^2 - 7x - 20 = (3x + 4)(2x - 5)$$

3. The critical values are

The critical values are

$$x = -\frac{4}{3} \quad \text{and} \quad x = \frac{5}{2}$$

4. Solve quadratic equation

Since the coefficient of x^2 is positive, the quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -\frac{4}{3}$ or $x > \frac{5}{2}$

PAST PAPER QUESTION

Q#: 21 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

21 *ABCD* is a square.

The point A has coordinates $(-5, 2)$
The point B has coordinates $(3, 5)$

Find an equation of the line that passes through B and C
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$A = (-5, 2), \qquad B = (3, 5)$$

2. The gradient of is

The gradient of AB is

$$\frac{5-2}{3-(-5)} = \frac{3}{8}$$

3. Find the gradient

Since $ABCD$ is a square, BC is perpendicular to AB.

4. Find the gradient

So the gradient of BC is

$$-\frac{8}{3}$$

5. The line through is

The line through $B = (3, 5)$ is

$$y - 5 = -\frac{8}{3}(x-3)$$

6. Multiply by 3

Multiply by 3:

$$3y - 15 = -8x + 24$$

$$8x + 3y - 39 = 0$$

FINAL ANSWER

$8x + 3y - 39 = 0$

PAST PAPER QUESTION

Q#: 21 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

21 *ABCD* is a square.

The point A has coordinates $(-5, 2)$
The point B has coordinates $(3, 5)$

Find an equation of the line that passes through B and C
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 **Marks: 4**

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$A = (-5, 2), \qquad B = (3, 5)$$

2. The gradient of is

The gradient of AB is

$$\frac{5-2}{3-(-5)} = \frac{3}{8}$$

3. Find the gradient

Since $ABCD$ is a square, BC is perpendicular to AB.

4. Find the gradient

So the gradient of BC is

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The line through $B = (3, 5)$ is

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6. Multiply by 3

Multiply by 3:

$$3y - 15 = -8x + 24$$

$$8x + 3y - 39 = 0$$

FINAL ANSWER
$8x + 3y - 39 = 0$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$
$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

2. Substitute

Substitute:

$$x^2 + (3x-1)^2 = 3x - 1 + 11$$

$$x^2 + 9x^2 - 6x + 1 = 3x + 10$$

$$10x^2 - 9x - 9 = 0$$

$$(5x+3)(2x-3) = 0$$

3. Evaluate fraction

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = \frac{3}{2}$$

4. Find

Find y:

$$x = -\frac{3}{5} \Rightarrow y = -\frac{14}{5}$$

$$x = \frac{3}{2} \Rightarrow y = \frac{7}{2}$$

FINAL ANSWER

$(x, y) = \left(-\frac{3}{5}, -\frac{14}{5}\right)$ or $\left(\frac{3}{2}, \frac{7}{2}\right)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$
$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

2. Substitute

Substitute:

$$x^2 + (3x - 1)^2 = 3x - 1 + 11$$

$$x^2 + 9x^2 - 6x + 1 = 3x + 10$$

$$10x^2 - 9x - 9 = 0$$

$$(5x + 3)(2x - 3) = 0$$

3. Evaluate fraction

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = \frac{3}{2}$$

4. Find

Find y:

$$x = -\frac{3}{5} \Rightarrow y = -\frac{14}{5}$$

$$x = \frac{3}{2} \Rightarrow y = \frac{7}{2}$$

FINAL ANSWER

$(x, y) = \left(-\frac{3}{5}, -\frac{14}{5}\right)$ or $\left(\frac{3}{2}, \frac{7}{2}\right)$

PAST PAPER QUESTION Q#: 23 Marks: 2

Topic: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

23 A curve has equation $y = \mathrm{f}(x)$
The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = \mathrm{f}(x) + 10$

(.................... , ...)
(1)

(ii) $y = \mathrm{f}(3x)$

(.................... , ...)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The minimum point of is

The minimum point of $y = f(x)$ is $(6, -3)$.

2. Calculate value

For

$$y = f(x) + 10$$

3. the graph moves 10 units

the graph moves 10 units up:

$$(6, -3) \to (6, 7)$$

4. Calculate value

For

$$y = f(3x)$$

5. the coordinate is divided by

the x-coordinate is divided by 3:

$$(6, -3) \to (2, -3)$$

FINAL ANSWER

$(6, 7)$ and $(2, -3)$

PAST PAPER QUESTION

Q#: 23 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

23 A curve has equation $y = f(x)$
The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 10$

(.................... , ...)
(1)

(ii) $y = f(3x)$

(.................... , ...)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The minimum point of is

The minimum point of $y = f(x)$ is $(6, -3)$.

2. Calculate value

For

$$y = f(x) + 10$$

3. the graph moves 10 units

the graph moves 10 units up:

$$(6, -3) \to (6, 7)$$

4. Calculate value

For

$$y = f(3x)$$

5. the coordinate is divided by

the x-coordinate is divided by 3:

$$(6, -3) \to (2, -3)$$

FINAL ANSWER

$(6, 7)$ and $(2, -3)$

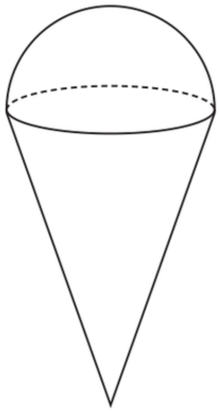
PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

24 The diagram shows a solid, **S**, made from a cone and a hemisphere.



The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of **S** is 4 × the radius of the hemisphere.

A separate sphere has radius kx cm.
The volume of this sphere is 12.5 × the volume of **S**

(a) Work out the value of k

k = ...

(4)

A solid, **T**, is similar to solid **S**

The volume of **T** is 512 × the volume of **S**

The total surface area of **T** is d × the total surface area of **S**

(b) Find the value of d

d = ...

(1)

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The hemisphere has radius

The hemisphere has radius x, so its height is x.

2. The total height of solid

The total height of solid S is $4x$, so the cone height is

$$4x - x = 3x$$

3. Volume of the cone

Volume of the cone:

$$\frac{1}{3}\pi x^2(3x) = \pi x^3$$

4. Volume of the hemisphere

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

5. Calculate volume

So the volume of solid S is

$$\pi x^3 + \frac{2}{3}\pi x^3 = \frac{5}{3}\pi x^3$$

6. Calculate volume

The sphere has radius kx, so its volume is

$$\frac{4}{3}\pi(kx)^3 = \frac{4}{3}\pi k^3 x^3$$

7. This is times the volume

This is 12.5 times the volume of S:

$$\frac{4}{3}\pi k^3 x^3 = 12.5\left(\frac{5}{3}\pi x^3\right)$$

8. Cancel

Cancel $\frac{1}{3}\pi x^3$:

$$4k^3 = 62.5$$

$$k^3 = 15.625 = \frac{125}{8}$$

$$k = 2.5$$

9. Calculate volume

For solid T, the volume scale factor is 512.

$$512 = 8^3$$

10. Calculate area

So the linear scale factor is 8, and the surface area scale factor is

$$8^2 = 64$$

FINAL ANSWER

$\boxed{k = 2.5,\ d = 64}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

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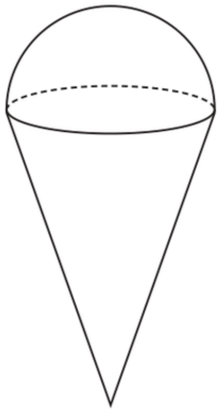

Diagram **NOT** accurately drawn

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The total height of **S** is 4 × the radius of the hemisphere.

A separate sphere has radius kx cm.
The volume of this sphere is 12.5 × the volume of **S**

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$k =$ ....

(4)

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The volume of **T** is 512 × the volume of **S**

The total surface area of **T** is d × the total surface area of **S**

(b) Find the value of d

$d =$ ....

(1)

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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Volume of the hemisphere:

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$$\pi x^3 + \frac{2}{3}\pi x^3 = \frac{5}{3}\pi x^3$$

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The sphere has radius kx, so its volume is

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This is 12.5 times the volume of S:

$$\frac{4}{3}\pi k^3 x^3 = 12.5\left(\frac{5}{3}\pi x^3\right)$$

8. Cancel

Cancel $\frac{1}{3}\pi x^3$:

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So the linear scale factor is 8, and the surface area scale factor is

$$8^2 = 64$$

FINAL ANSWER

$\boxed{k = 2.5,\ d = 64}$

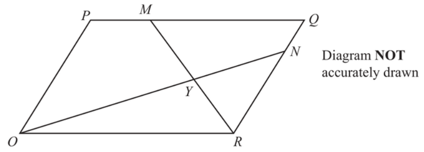
PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

25 *OPQR* is a parallelogram.



$\overrightarrow{OP} = 2\mathbf{a}$ and $\overrightarrow{OR} = 3\mathbf{b}$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of **a** and **b**, giving your answers in simplest form

(i) $\overrightarrow{ON}$

....

(1)

(ii) $\overrightarrow{MR}$

....

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$k =$ ....

(4)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION *Dr. Eslam Ahmed*

1. Calculate value

$$\overrightarrow{OP} = 2a, \qquad \overrightarrow{OR} = 3b$$

2. Calculate value

So

$$\overrightarrow{OQ} = 2a + 3b$$

3. Evaluate fraction

Since $PM : PQ = 1 : 4$,

$$\overrightarrow{PM} = \frac{1}{4}\overrightarrow{PQ}$$

$$\overrightarrow{PM} = \frac{1}{4}(3b) = \frac{3}{4}b$$

$$\overrightarrow{OM} = 2a + \frac{3}{4}b$$

4. Evaluate fraction

Since $RN : RQ = 4 : 5$,

$$\overrightarrow{RN} = \frac{4}{5}\overrightarrow{RQ}$$

$$\overrightarrow{RN} = \frac{4}{5}(2a) = \frac{8}{5}a$$

$$\overrightarrow{ON} = 3b + \frac{8}{5}a$$

5. Evaluate fraction

So

$$\overrightarrow{MR} = \overrightarrow{OR} - \overrightarrow{OM}$$

$$\overrightarrow{MR} = 3b - \left(2a + \frac{3}{4}b\right)$$

$$\overrightarrow{MR} = -2a + \frac{9}{4}b$$

6. Let be on

Let Y be on ON, so

$$\overrightarrow{OY} = k\overrightarrow{ON} = k\left(\frac{8}{5}a + 3b\right)$$

7. Evaluate fraction

Also Y is on MR, so

$$\overrightarrow{OY} = \overrightarrow{OM} + t\overrightarrow{MR}$$

$$\overrightarrow{OY} = 2a + \frac{3}{4}b + t\left(-2a + \frac{9}{4}b\right)$$

8. Evaluate fraction

Compare coefficients of a and b:

$$\frac{8}{5}k = 2 - 2t$$

$$3k = \frac{3}{4} + \frac{9}{4}t$$

9. Solve equation

From the first equation,

$$t = 1 - \frac{4}{5}k$$

10. Substitute into the second equation

Substitute into the second equation:

$$3k = \frac{3}{4} + \frac{9}{4}\left(1 - \frac{4}{5}k\right)$$

$$3k = 3 - \frac{9}{5}k$$

$$\frac{24}{5}k = 3$$

$$k = \frac{5}{8}$$

FINAL ANSWER

$\overrightarrow{ON} = \frac{8}{5}a + 3b$, $\overrightarrow{MR} = -2a + \frac{9}{4}b$, and $k = \frac{5}{8}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

25 *OPQR* is a parallelogram.

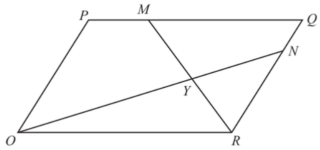


Diagram **NOT** accurately drawn

$\overrightarrow{OP} = 2\mathbf{a}$ and $\overrightarrow{OR} = 3\mathbf{b}$

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The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of **a** and **b**, giving your answers in simplest form

(i) $\overrightarrow{ON}$

....

(1)

(ii) $\overrightarrow{MR}$

....

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$k =$ ....

(4)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$\overrightarrow{OP} = 2a, \qquad \overrightarrow{OR} = 3b$$

2. Calculate value

So

$$\overrightarrow{OQ} = 2a + 3b$$

3. Evaluate fraction

Since $PM : PQ = 1 : 4$,

$$\overrightarrow{PM} = \frac{1}{4}\overrightarrow{PQ}$$

$$\overrightarrow{PM} = \frac{1}{4}(3b) = \frac{3}{4}b$$

$$\overrightarrow{OM} = 2a + \frac{3}{4}b$$

4. Evaluate fraction

Since $RN : RQ = 4 : 5$,

$$\overrightarrow{RN} = \frac{4}{5}\overrightarrow{RQ}$$

$$\overrightarrow{RN} = \frac{4}{5}(2a) = \frac{8}{5}a$$

$$\overrightarrow{ON} = 3b + \frac{8}{5}a$$

5. Evaluate fraction

So

$$\overrightarrow{MR} = \overrightarrow{OR} - \overrightarrow{OM}$$

$$\overrightarrow{MR} = 3b - \left(2a + \frac{3}{4}b\right)$$

$$\overrightarrow{MR} = -2a + \frac{9}{4}b$$

6. Let be on

Let Y be on ON, so

$$\overrightarrow{OY} = k\overrightarrow{ON} = k\left(\frac{8}{5}a + 3b\right)$$

7. Evaluate fraction

Also Y is on MR, so

$$\overrightarrow{OY} = \overrightarrow{OM} + t\overrightarrow{MR}$$

$$\overrightarrow{OY} = 2a + \frac{3}{4}b + t\left(-2a + \frac{9}{4}b\right)$$

8. Evaluate fraction

Compare coefficients of a and b:

$$\frac{8}{5}k = 2 - 2t$$

$$3k = \frac{3}{4} + \frac{9}{4}t$$

9. Solve equation

From the first equation,

$$t = 1 - \frac{4}{5}k$$

10. Substitute into the second equation

Substitute into the second equation:

$$3k = \frac{3}{4} + \frac{9}{4}\left(1 - \frac{4}{5}k\right)$$

$$3k = 3 - \frac{9}{5}k$$

$$\frac{24}{5}k = 3$$

$$k = \frac{5}{8}$$

FINAL ANSWER

$\overrightarrow{ON} = \frac{8}{5}a + 3b$, $\overrightarrow{MR} = -2a + \frac{9}{4}b$, and $k = \frac{5}{8}$

PAST PAPER QUESTION Q#: 26 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

26 Write $4 - \left[(3x-5) \div \frac{3x^2+x-10}{4x-1} \right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION Q#: 26 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

$$4-\left[(3x-5)\div\frac{3x^2+x-10}{4x-1}\right]$$

2. Factor the quadratic

Factor the quadratic:

$$3x^2+x-10=(3x-5)(x+2)$$

3. Use trigonometry

So the expression inside the square brackets is

$$(3x-5)\div\frac{(3x-5)(x+2)}{4x-1}$$

4. Dividing by a fraction means

Dividing by a fraction means multiplying by its reciprocal:

$$(3x-5)\times\frac{4x-1}{(3x-5)(x+2)}$$

5. Cancel

Cancel $3x-5$:

$$\frac{4x-1}{x+2}$$

6. Use trigonometry

Therefore the full expression is

$$4-\frac{4x-1}{x+2}$$

7. Use the common denominator

Use the common denominator $x+2$:

$$\frac{4(x+2)-(4x-1)}{x+2}$$

$$=\frac{4x+8-4x+1}{x+2}$$

$$=\frac{9}{x+2}$$

FINAL ANSWER

$$\boxed{\frac{9}{x+2}}$$

PAST PAPER QUESTION

Q#: 26 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

26 Write $4 - \left[(3x-5) \div \dfrac{3x^2 + x - 10}{4x - 1}\right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

$$4 - \left[(3x-5) \div \frac{3x^2+x-10}{4x-1}\right]$$

2. Factor the quadratic

Factor the quadratic:

$$3x^2+x-10 = (3x-5)(x+2)$$

3. Use trigonometry

So the expression inside the square brackets is

$$(3x-5) \div \frac{(3x-5)(x+2)}{4x-1}$$

4. Dividing by a fraction means

Dividing by a fraction means multiplying by its reciprocal:

$$(3x-5) \times \frac{4x-1}{(3x-5)(x+2)}$$

5. Cancel

Cancel $3x-5$:

$$\frac{4x-1}{x+2}$$

6. Use trigonometry

Therefore the full expression is

$$4 - \frac{4x-1}{x+2}$$

7. Use the common denominator

Use the common denominator $x+2$:

$$\frac{4(x+2)-(4x-1)}{x+2}$$

$$= \frac{4x+8-4x+1}{x+2}$$

$$= \frac{9}{x+2}$$

FINAL ANSWER

$$\boxed{\frac{9}{x+2}}$$